BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Images, Pixels and File Sizes

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

SILENT

BLACK PEN: Answer all four before looking at the model answers.

1. Previous lesson: which standard supports many writing systems?

2. Convert 48 bits into bytes.

3. Convert binary 00010101 to denary.

4. 150 characters use eight bits each. Find bits and bytes.

Today's objectives

Explain pixels and colour depth; calculate raw bitmap size in bits and bytes.

Key vocabulary - Read, cover, recall

pixel One bitmap picture element	**resolution** Width and height in pixels
colour depth Bits for each pixel's colour	**metadata** Information about a file

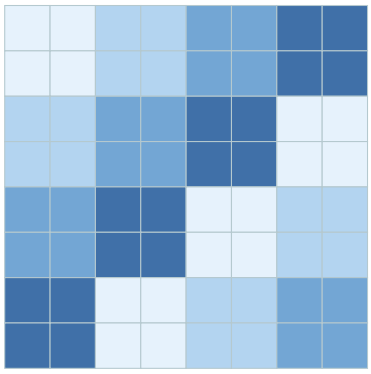
Write one definition in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Pixels, resolution and depth

A bitmap is a grid of pixels. Resolution is width x height in pixels.



8 x 8 = 64 pixels

Each square is one pixel.

Colour depth is bits per pixel.

More pixels or more bits usually means more raw image data.

I Do - Raw bitmap file-size method

LISTEN

Paper 1 guide example: 8 x 8 bitmap, 2-bit colour depth.

WIDTH x HEIGHT x COLOUR DEPTH = BITS

8 x 8 = 64 pixels; 64 x 2 = 128 bits.

128 / 8 = 16 bytes (uncompressed pixel data only).

Headers, metadata and compression can change actual file size.

We Do - Guided practice

MWBs

10 x 6 pixels at 4 bits each: find pixels, bits and bytes.

Pixels: _____ x _____ = _____

Bits: _____ x 4 = _____

Bytes: _____ / 8 = _____

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

MWBs

Start on paper. Do not look at model answers until page 4.

```
width = 10    height = 8    depth = 4
print(width * height * depth)
```

Predict the number of bits printed by the starter.

Change depth to 8. Predict the new bits.

Add a print expression to display bytes, not bits.

You Do - Independent work

SILENT

Define pixel and colour depth. [2]

A 12 x 5 bitmap has 4-bit depth. Find bits and bytes. [3]

A 1000 x 800 bitmap has 8-bit depth. Find raw MB. [4]

Thinking: Why might the real file size differ?

Exit Ticket - True or False?

THUMBS

Answer before you turn to the green-pen page.

1. Image resolution describes the number of pixels. ☐ T ☐ F
2. Doubling only colour depth doubles raw pixel data. ☐ T ☐ F
3. The raw pixel calculation includes metadata. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick correct answers. For a mistake, use the model answer then copy the FULL question and write a fresh answer below.

Do Now - Answers

Q1 Unicode ☐

Q2 48 / 8 = 6 bytes ☐

Q3 16 + 4 + 1 = 21 ☐

Q4 1200 bits = 150 bytes ☐

We Do - Guided practice

60 pixels; 240 bits; 30 bytes

Key vocabulary

pixel ☐
One bitmap picture element

resolution ☐
Width and height in pixels

colour depth ☐
Bits for each pixel's colour

metadata ☐
Information about a file

Code First - Answers

GREEN PEN

Predict ☐

10 x 8 x 4 = 320 bits

Write ☐

10 x 8 x 8 = 640 bits

MWBs

Improve ☐

print(width * height * depth / 8) = 80 bytes

Exit Ticket - Explanations

1. True

2. True if dimensions do not change

3. False: metadata is extra

You Do - Self-mark

GREEN PEN

Q1	A pixel is an individual element; depth is bits per pixel	☐
Q2	12 x 5 x 4 = 240 bits = 30 bytes	☐
Q3	6,400,000 bits = 800,000 bytes = 0.8 MB	☐
Thinking	Headers, metadata or compression affect actual size.	☐

Try the question again

GREEN PEN

1. Copy the FULL question you got wrong (or the hardest):

2. In GREEN PEN, write a NEW answer. Show working: